

ShiftTouch: Sheet-type Interface Extending Capacitive Touch Inputs with Minimal Screen Occlusion

Kaori Ikematsu
Yahoo Japan Corporation
Tokyo, Japan

Kunihiro Kato
Tokyo University of Technology
Tokyo, Japan

Rei Kawakami
Tokyo University of Technology
Tokyo, Japan

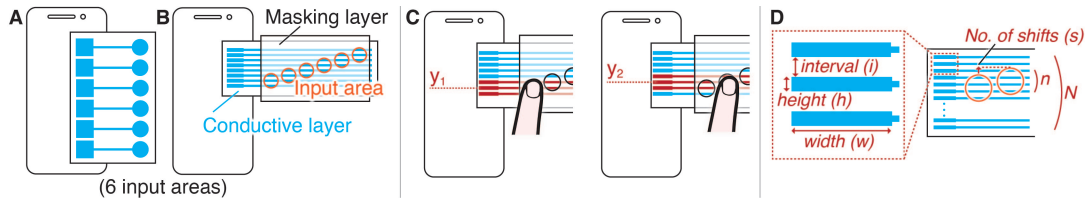


Figure 1: (A) A typical touch extension with conductive patterns. (B) Basic structure of ShiftTouch. (C) Touch surface can identify touch inputs from each input area by a slight shift in touch coordinates. (D) Parameters for conductive layer.

ABSTRACT

We present ShiftTouch, a sheet-type passive interface that provides multiple inputs for capacitive touch surfaces with minimal screen occlusion. It consists of a conductive layer and a masking layer that partially insulates the conductive one. ShiftTouch uses multiple linear electrodes to control the fine touch position. The touch input is triggered under the electrodes when several adjacent electrodes are grounded simultaneously. Although each input area shares some electrodes with neighboring input areas, the touch surface can identify the inputs from each input area by detecting the slight displacement of the touch position. Our interface is simple yet effective in implementing multiple input areas while reducing screen occlusion compared to existing approaches using finger-sized electrodes.

CCS CONCEPTS

• Human-centered computing → Interaction techniques.

KEYWORDS

Capacitive Touch Surface; Conductive Ink; Printed Interface.

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1 INTRODUCTION

In recent years, capacitive touch surfaces such as smartphones, smartwatches, and tablet PCs have been widely used. Although

touch surfaces were originally designed to detect finger touches, they can also detect grounded conductive objects other than fingers. Various passive interfaces using conductive materials such as metal, conductive inks, and rubber have been proposed to enhance the expressiveness of touch interaction. These are useful in various applications, such as expanding the input area of touch surfaces [1, 3, 10], adding tactile feedback [6, 11], and establishing communication channels between multiple devices [4]. However, a conductive object with an area the size of a fingertip, as shown in Fig. 1 (A), needs to contact the touch surface to trigger a touch event. Thus, if many conductive parts are attached to the screen, screen occlusion will be a serious problem.

To address this, we propose ShiftTouch, a sheet-type passive interface that provides multiple input areas with minimal screen occlusion. As shown in Fig. 1 (B), the electrode of the interface is divided into multiple thin lines that are connected to a conductive pattern extending outside the screen. We installed a masking layer on the conductive layer to partially expose the conductive pattern. Although each input area shares some electrodes with neighboring input areas, the touch surface can identify the inputs from each input area by detecting the slight displacement of the touch position (Fig. 1 (C)). Our interface is simple yet effective in implementing multiple input areas while reducing screen occlusion compared to existing approaches using finger-sized electrodes.

2 INTERFACE STRUCTURE

Figure 1 (B) shows the basic structure of ShiftTouch. It consists of a conductive layer and an insulation layer that masks the conductive one. On the conductive layer, linear electrodes and conductive patterns extending from the electrodes are printed with silver nanoparticle ink. When a user touches the conductive pattern, a certain number of electrodes are grounded, and touch input is generated underneath the electrodes. The center coordinate of the touch input detected by the touch driver is generally located near the center of the grounded electrode area. Although commercially available touch surfaces use about 5 to 8-mm width touch electrodes [5, 9], the touch driver can detect a touch position more accurately than millimeter-scale because the driver complements the touch position.

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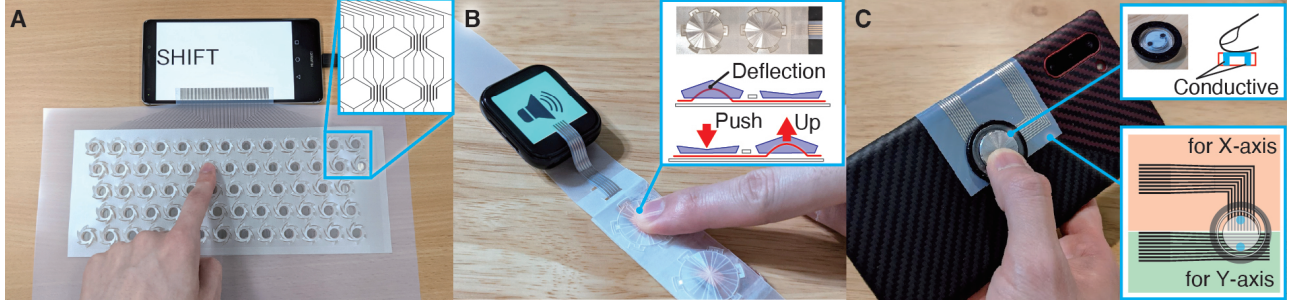


Figure 2: (A) Keyboard application with 63 input areas. Add-on sheet with Kirigami buttons is attached on the conductive layer. (B) Seesaw switch-like add-on embedded in the smartwatch band. (C) Joystick application for detecting the 2D operations.

Table 1: A comparison of occluded areas for a keyboard with 63 keys implemented by each method. For (a) and (b), the electrodes are square, thus $w = h$.

	ShiftTouch			(a) Single-layer			(b) Double-layer [1, 10]		
w (mm)	6	8	10	6	8	10	6	8	10
Occluded area (mm^2)	446	594	743	2,417	4,230	6,548	612	1,072	1,660

Thus, even a shift of a few electrodes to be grounded is sufficient to cause a change in the touch position to identify inputs from each input area (Fig. 1 (C)). It is known that linear electrodes can be used to perform continuous touch operations such as scrolling by moving a finger on a conductive pattern [7]. In ShiftTouch, we utilized this to control the slight displacement of the touch position, rather than tracking continuous finger movements. However, it is difficult to precisely touch only the part connected to optional electrodes in a dense conductive pattern with a finger. Therefore, a masking layer with holes that partially expose the conductive pattern was installed to design an input part so that the user does not touch the unnecessary part.

With our interface, when the electrode height is h , the interval of each electrode is i , and the electrode is shifted by s (Fig. 1 (D)), the threshold for distinguishing inputs from adjacent input areas (i.e., acceptable error from the center coordinate of the electrode) can be expressed as $s \cdot (i + h)/2$. When the total number of electrodes is N , and the number of electrodes used for each input is n , the feasible number of inputs can be expressed as $\lfloor (N - n)/s + 1 \rfloor$ with the conditions of $N > n$, $N - n \geq s$, and $s > 0$.

3 APPLICATION

In ShiftTouch, various add-ons can be attached on the basic structure to extend interactions of the touch surface. Figure 2 (A) shows the conductive pattern for a keyboard layout with an add-on sheet featuring Kirigami buttons. We leverage the geometric pattern of the Rotational Erection System [8] for the button design. It has 63 keys and functions as a QWERTY keyboard when the electrodes are attached to smartphones or tablet PCs.

Figure 2 (B) shows the ShiftTouch with a seesaw switch-like add-on connected to a smartwatch. This button is based on a split-fold waterbomb [2] shape and has bistable properties. A loosely flexed PET film is inserted between the top layer and the conductive layer so that when one button is pressed, the deflection moves under the other button, pushing the button up. The PET film also has a conductive pattern, where one end of the pattern is connected to the device casing, and the other end is connected to the conductive

pattern underneath either of the buttons. This allows explicit mode switching for smartwatch operations.

Figure 2 (C) shows the joystick-type add-on for back-of-device interactions. A disk-shaped input part with two conductive points is used in this interface instead of a masking layer with holes. Under the input part, two separate conductive patterns are installed. One is used to detect the displacement of the input part in the x-axis direction and the other that in the y-axis direction. When the user moves the input part, the electrode to be connected is changed, so multiple levels of displacement can be detected in each axial direction. We used the parameters of $w = 10$ mm, $h = 0.6$ mm, and $i = 0.5$ mm (the optimal parameters for [7]) for the prototypes.

4 DISCUSSION

To assess how effective ShiftTouch is in reducing the occluded area, we compared it with (a) a typical interface having a one-to-one correspondence between the input area and the electrodes (Fig. 1 (A)) and (b) an interface having a two-layer matrix structure of conductive patterns (touching an input area will trigger two touch events on the touch surface), as in [1, 10]. We define the occluded area as a rectangular area containing all the electrodes on the touch surface, specifically, $(N \times h + (N - 1) \times i) \times w$.

Here, we take the example of the interface with 63 inputs shown in Fig. 2 (A). We implemented the ShiftTouch keyboard with 68 electrodes and parameters of $s = 1$ and $n = 6$. The parameters of electrodes are the same as those used in the experiment ($h = 0.6$ and $i = 0.5$), so the occluded area is $74.3w \text{ mm}^2$. To achieve 63 inputs, (a) requires 63 electrodes and (b) requires 16 electrodes (eight in each layer). We calculated the occlusion area as containing electrodes that are all square, with sides of 6 mm, 8 mm, and 10 mm, and an interval (i) of 0.4 mm (the parameter used in [10]). The results shown in Table 1 indicate that ShiftTouch can effectively reduce the screen occlusion under the same w condition.

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